

# HERMANN PAUWELLS RIVERA

## Full-Stack Developer | Software Engineer

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### PROFILE

Full-stack developer specialized in TypeScript, React/Next.js and Node.js, graduating soon from a Computer Science Engineering degree at Tecnológico de Monterrey. Experienced building software for real businesses and users, including e-commerce, real-time systems and APIs. Extra experience in C++/JUCE, DSP and machine learning.

### TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, C++, SQL, Swift, C#  
**Frontend:** React, Next.js, Vite, Tailwind CSS, Three.js  
**Backend & Data:** Node.js, Express, Supabase, PostgreSQL, MySQL, FastAPI  
**Tools:** Git, GitHub, GitHub Actions, CI, JUCE, TensorFlow, Unity

### EXPERIENCE

**Full-Stack Developer — Juguetería El Arbolito** 2026–present

- Developing and implementing an e-commerce and admin platform for a single-branch retail business (catalog, cart, checkout, backend).
- Mercado Pago payments and inventory sync between online sales and point-of-sale (local and online).

### SELECTED PROJECTS

**Lumina Reservations (WorkHub MTY)** Academic team project · Node/Express backend, real-time (SSE) and frontend 2026

- Implemented the reservation backend in Node/Express (services, repositories, controllers, migrations) and real-time occupancy updates over SSE.
- Built the floor map and the reservation flow in React/Vite (dashboard, filters, confirmation) with role controls.

**folk-park** Personal project 2026

- Wavetable VST3/Standalone synthesizer in C++20 and JUCE with deterministic MIDI generation.
- DSP architecture and automated tests; validated with pluginval strictness 5.

**HeatShield** iOS app 2025

**Winner, Sustainability category, Swift Challenge Fest 2025 (Tecnológico de Monterrey)**

- iOS app with real-time heat alerts (OpenWeatherMap), heat index and shelter maps.

Other projects: CrediFiel Challenge (Datathon 2025) · Multi-agent warehouse simulation (RL) · MAWI · VisualControl (IoT) · Mildred Pierce · GateGenius (HackMTY 2025).

### EDUCATION

**Computer Science Engineering (ITC) — Tecnológico de Monterrey, Monterrey Campus — 2021–2026 · Graduation expected: December 2026**

### ACTIVITIES

- SwiftTec Coding Community (2025) — hackathons and global innovation challenges.
- Music production and sound design (FL Studio).

Spanish — native · English — C1